

Manar Attar

AI Researcher & NLP / LLM Engineer

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Summary

AI and NLP engineer who ships production AI systems end-to-end — from fine-tuned transformer models to agentic RAG pipelines and the interfaces around them. Published researcher (ACM SIGSPATIAL) with an MA in Language & AI from Vrije Universiteit Amsterdam. Specialises in agentic architectures, LLM tool-calling, and taking AI ideas from prototype to deployed product — equally comfortable training a model, wiring up a backend, and building the interface around it.

Technical Skills

Programming: Python, SQL, JavaScript, R

LLM & NLP: OpenAI API, GPT-4o & GPT-4o-mini, Ollama (Llama 3.1, Qwen3), Hugging Face Transformers (BERT, RoBERTa, HateBERT), RAG, ChromaDB, Prompt Engineering, Agentic & Multi-Agent Systems, MCP, Whisper (STT), ElevenLabs (TTS)

ML & Data: PyTorch, scikit-learn, pandas, NumPy, Matplotlib, seaborn

Web & Infra: FastAPI, Flask, SQLAlchemy, Pydantic, REST & SSE APIs, PostgreSQL, Firebase, Docker, React, Vite, Tailwind CSS, Git, Vercel, Render

Languages: English (Fluent), Arabic (Native), Dutch (B2)

Work Experience

AI Consultant | Vervai

Apr 2024 – Present

- Deliver end-to-end AI solutions across client engagements — custom LLM applications, conversational assistants, and RAG systems over proprietary document collections, alongside workflow automation.
- Advise on AI strategy, scoping high-impact use cases and guiding clients from concept to production.
- Lead hands-on AI workshops that upskill client teams; shipped **SwipeEat**, a swipe-based meal-recommendation MVP, to production.

Innovation Consultant | Daffee

Jun 2024 – Sep 2025

- Built **4 role-specific GPT assistants** (marketing, sales, operations, and leadership) that accelerated daily workflows and decision-making.
- Designed semi-automated workflows that cut repetitive manual work and freed teams for higher-value tasks.

Education

MA Language & AI | Vrije Universiteit Amsterdam

Sep 2025 – Jul 2026

- Specialisation in NLP, large language models, data annotation, and cross-dataset evaluation.

Amsterdam Startup Launch Program | Vrije Universiteit Amsterdam

Sep 2024 – Jan 2025

- Intensive entrepreneurship program: validating business concepts with the Business Model Canvas.

BSc Artificial Intelligence | Vrije Universiteit Amsterdam

Sep 2020 – Aug 2023

- Foundations of AI, machine learning, statistics, and hypothesis testing. Minor in Data Science (NLP & text analysis).

Master's Thesis

Author Profiling of Hate Speech Spreaders: Zero-shot LLMs vs. Fine-tuned Encoders

MA Thesis, 2026

- Investigated whether the age and gender of hate speech authors can be inferred from their writing, comparing two competing paradigms under identical conditions: **zero-shot large language models** (LLaMA-3.1-8B, Qwen3-32B) prompted without task-specific training, and **encoder models fine-tuned** on labelled data (BERT, HateBERT, RoBERTa), with an **SVM** lexical baseline for reference.
- Designed a cross-domain evaluation — models trained on **PAN14** social media data and tested on **LiLaH-HAG** hate speech comments — to measure how well author profiling transfers into an abusive-language domain it was not built for.
- Found that **neither approach reached performance sufficient to recommend it** for deployment: zero-shot LLMs held a narrow edge on gender and fine-tuned encoders on age, but all results sat close to the majority baseline, so the comparison is inconclusive by design rather than by omission.
- Argued from confusion-matrix and lexical error analysis that the ceiling is a property of the domain, not the models: hate speech is a strongly conventionalised register in which shared genre norms suppress the individual stylistic signal author profiling depends on.
- Extended the study to **Slovene** (CroSloEngual BERT) to test whether grammatical-gender marking supplies a signal English lacks — the strongest result in the study, and evidence that language-specific cues matter more than model scale.

AI Projects

TelecomNL Voice AI Assistant

- Built a bilingual (NL/EN) voice agent combining Whisper speech-to-text, GPT-4o reasoning, and ElevenLabs speech synthesis, handling **10 telecom issue categories** across **8 customer profiles**.
- Added a live support dashboard tracking intent, sentiment, and confidence, with rule- and LLM-based escalation and auto-generated handoff summaries.

Skills: FastAPI, GPT-4o, ElevenLabs Conversational AI, WebSocket, Whisper, JavaScript.

AI Contract Risk Analyzer

- Built a RAG assistant with **per-clause risk scoring (0–100)** and proposed revisions across PDF, DOCX, and TXT uploads up to 10 MB, flagging missing standard clauses.
- Added source-grounded document Q&A and one-click PDF report export.

Skills: FastAPI, ChromaDB, OpenAI, RAG, React.

Multi-Agent Research Assistant

- Engineered a **5-agent** pipeline (coordinator, search, analysis, fact-check, synthesis) that produces structured, citation-backed research reports, with each agent's progress streamed live to the interface.
- Implemented follow-up Q&A grounded in the sources of the generated report.

Skills: Agentic AI, FastAPI, OpenAI, Tavily, SSE, React.

Munazara: AI Debate Engine

- Designed a live-deployed debate platform where **3 agents (PRO / CON / Judge)** argue across **4 structured rounds**, backed by RAG-sourced evidence and real-time fact-checking.
- Added a user-vs-AI mode, before/after voting, and an interactive knowledge graph linking every debate.

Skills: FastAPI, gpt-4o-mini, Tavily, ChromaDB, React, SSE.

RivalScan: Competitor Intelligence Dashboard

- Automated competitor tracking across three sources (RSS feeds, GitHub releases, and changelogs) in a single dashboard.
- Summarised each update with AI and scored it by business impact to speed prioritisation.

Skills: FastAPI, OpenAI, SQLAlchemy, React, Vite.

SwipeEat: Adaptive Meal Recommendation

- Implemented a preference-based recommendation algorithm matching users to meals from swipe feedback.
- Designed a swipe-driven mobile-web interface and delivered it as a working MVP for Vervai.

Skills: Python, Flask, JavaScript, HTML, CSS.

Academic Projects

LINKED4RESILIENCE: Geo-Data Integration for Crisis Response

- Built data pipelines for cleaning, processing, and visualising geo-annotated crisis datasets.
- Applied Linked Data methodologies and implemented an algorithm for data unification and integration.

Result: Published at the 31st ACM SIGSPATIAL International Conference (2023).

Skills: Python, RDF, SPARQL, Ontology, Linked Data, Geodata, data visualisation.

Emotion & Sentiment Analysis (Human Language Technology)

- Developed sentiment and emotion classification models with optimised SVMs on conversational and social-media text.
- Applied TF-IDF feature engineering and error analysis to study trade-offs under class imbalance.

Result: Competitive emotion classification under imbalanced datasets.

Skills: Python, NLP, SVM, Text Mining, scikit-learn.

Publications

• **M. Attar**, S. Wang, R. Siebes, E. Kultorp (2023). *Converting and Enriching Geoannotated Event Data: Integrating Information for Ukraine Resilience*. 31st ACM SIGSPATIAL International Conference, Nov 13–16, 2023.

• **M. Attar**, S. Wang, R. Siebes, E. Kultorp (2023). *Using Integrated and Enriched Linked Data for Ukraine Resilience*. BNAIC 2023 Conference, Nov 8–10, 2023.

References

Dr. Ilia Markov — Master's Thesis Supervisor

Assistant Professor, Computational Linguistics & Text Mining Lab (CLTL), Vrije Universiteit Amsterdam

Raihan Karim — Business Partner & Academic Collaborator

Long-standing collaborator across academic projects and entrepreneurial ventures

Contact details available on request.